RICHFIELD GRADUATE INSTITUTE OF TECHNOLOGY

IT PROJECT 700 PHASE 2

Group members

STUDENT NO.	NAME	SURNAME	EMAIL	CONTACT NO.
402308936	Tristan	Moses	402308936@my.richfield.ac.za	0828015082
402411668	Reenesh	Ramawthar	402411668@my.richfield.ac.za	0846041521
402411227	Joyce	Kamalu	402411227@my.richfield.ac.za	0671075365
402412304	Novithan	Naidoo	402412304@my.richfield.ac.za	0620829186
402413409	Thingo	Shezi	402413409@my.richfield.ac.za	0714299251
402111846	Lungelo	Ngwane	402111846@my.richfield.ac.za	0642534563

IDENTIFICATION OF NEED

The problem that was identified was that Richfield has eight campuses across four provinces and as private higher education provider they do not own any accommodation houses because direct review of richfield.ac.za and its published FAQ confirmed that campuses currently refer students to off-campus housing informally, leaving students with no centralized way to search, compare prices, check distances or verify if a place has been inspected. With the off-campus housing there is no visibility for the parents or guardians who pay because they only see the safety, deposit and lease terms only after an informal decision has been made. Another issue that was noted was that the funding is often misunderstood because of Richfield's private status, meaning that its students are not NSFAS eligible since NSFAS only funds public universities and TVET Colleges. The solution was to build a single, campus-linked, verified accommodation search and booking system to replace informal word-of-mouth referrals.

PRELIMINARY INVESTIGATION

The research conducted was to make sure the project resembles an extension of Richfield College and through the investigation we:

- Checked Richfield's site for real brand colors, fonts and campus locations.
- Analyzed standard rental sites to list essential features such as budget filters, distance to the campuses, shortlisting and inquiry forms.
- Cross-checked NSFAS rules to confirm private university students cannot claim NSFAS accommodation funding.
- Assessed team member skill sets (PHP, HTML, CSS, JS, MySQL) and machine setups (XAMPP).

FEASIBILITY STUDY

TECHNICAL

- The required skills for this project are PHP, HTML, CSS, JS, MySQL which were already within the teams' available skills set.
- Leaflet + OpenStreetMap gives the mapping with no API key and no billing account unlike Google Map and this removes a live demo failure point.
- The risk found was that a machines existing MySQL can already occupy port 3306 and mitigation for this would re-pointing XAMPP's MariaDB to port 3307.

OPERATIONAL

- The interface degrades gracefully and with no database configured it runs on built-in sample data so that any team member or marker can view it with zero setup.
- No end-user training will be needed because search, filter, enquiry match are from everyday website use-cases from property and e-commerce websites.

ECONOMIC

- Every component in the project, PHP, Apache, the database, Leaflet, Google fonts are all free and open source, there were no licenses or API billing at any stage.
- Cost is limited because beyond coursework, ordinary shared PHP/MySQL hosting is sufficient and needs no specialized infrastructure.

PROJECT PLANNING

Project planning includes scope, prioritization and the constraints carried through every decision.

When using the MoSCoW scope for this we find that with:

- **Must:** There needs to be a landing page, campus search, listing details, viewing-request form and a support database.
- **Should:** It should have an accessible and responsive design
- **Could:** Have admin CRUD for listing, student login and persistent server-side shortlist.
- **Won't (Yet):** have a payment processing and live third-party booking integrations.

CONSTRAINTS DESIGNED IN FROM THE START

- This system was developed as an isolated module inside the groups shared repository, so that it could not conflict with the parallel work already under way on the wider emergex github.
- This system uses the institutions real name and branding without Richfield's endorsement so a visible sample data notice of "not affiliated" footer disclaimer was planned in from day one and was not added as an afterthought.
- There was no project budget because every planning decision be it was the stack, mapping library and hosting, was constrained to free and open-source options.

PROJECT SCHEDULING

THE TESTING SCOPE AND METHODOLOGY (Gantt and PERT Chart)

- **Testing types covered**: Regression, UI, End-to-end, Integration, Security and functional testing
- **Test automation and Frameworks**: Automated UI, Functional, Integration and security testing suites are executed using playwright and NUnit frameworks to validate user workflows, form security and responsive layouts.

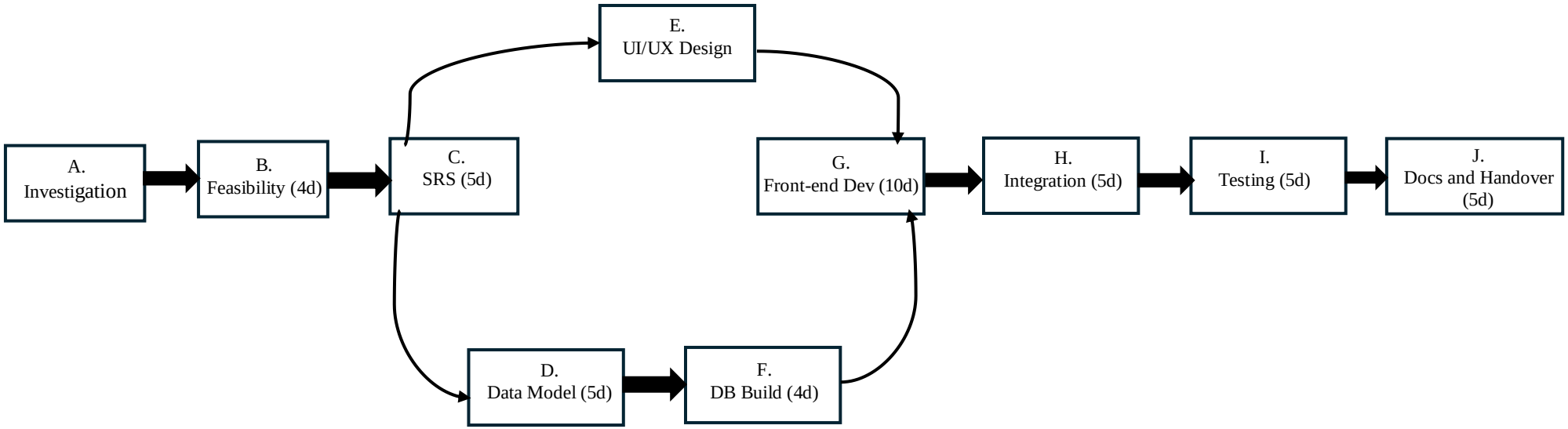
Gantt Chart

The below ten activities, sequenced with dependencies run to a 47 day working day critical path and both these charts below share the same schedule

Phase	Task	01-07 Sep	08-14 Sep	15-21 Sep	22-28 Sep	29 Sep-05 Oct	06-12 Oct	13-19 Oct	20-26 Oct	27 Oct-02 Nov	03-09 Nov
Planning	Identification and Investigation										
	Feasibility Study										
	Requirements Gathering										
Design	Data Modelling and DB Design										
	UI / UX Design										
Implementation	Database Implementation										
	Front-end Development										
Integration and Close-out	System Integration										
	Testing										
	Documentation and Handover										

Task duration UI / UX Design Current date marker

PERT CHART



Testing Deliverables and Metric Summary

- **Written Test Cases**: 36 manually authored test cases covering functional requirements FR1 through FR10
- **Automated Test Executions:** 118 Test cases executed against the live demo platform via playwright and NUnit

Execution status: Deno website testing phase was completed and the final acceptance test run scheduled upon delivery of the final production build

SOFTWARE REQUIREMENT SPECIFICATION

The Software Requirement Specification acts as the primary operational blueprint for quality assurance and system verification. While development team members construct system behavior around the database data models and backend PHP scripts, testing activities are driven strictly from the functional (FR1-FR10) and non-functional (NFR1-NFR6) requirements defined below.

FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT
FR1	Display every campus with city, province, confirmed/unconfirmed status and bed count.
FR2	Let a student search listings by campus, maximum rent, room type and required amenities.
FR3	Show rent, room type, verified/inspected status, amenities and walking/driving distance to campus for each listing.
FR4	Plot listings and their campus together on an interactive map.
FR5	Let a student shortlist up to four listings, persisting across reloads.
FR6	Accept a viewing request: contact details, preferred date, move-in month, selected listings.
FR7	Validate all viewing-request input server-side, including South African mobile format, before accepting it.
FR8	Estimate total move-in cost including deposit, admin fee, rent for a chosen lease length and payment plan.

ID	REQUIREMENT
FR9	Protect the viewing-request form against cross-site request forgery.
FR10	Continue showing campus, listing and FAQ content correctly with no database connected, using built-in sample data.

Non-functional requirements

ID	QUALITY	REQUIREMENT
NFR1	Performance	Pages render server-side with no client build step, over ordinary shared hosting.
NFR2	Security	Every database query uses a parameterized statement; user input is never concatenated into SQL.
NFR3	Portability	Runs unmodified against MySQL/MariaDB or SQLite via PHP's PDO layer.
NFR4	Accessibility	Meets WCAG 2.2 AA color contrast and every control is keyboard-operable.
NFR5	Honesty	Never presents sample data as real availability, and never claims NSFAS eligibility or an unheld affiliation.
NFR6	Compatibility	No horizontal overflow from 360px to 1920px viewport width.

DATA MODELS

The below nine table references the data that is maintained by the staff and the data captured by the students using the system.

